

# Vatsala Swaroop

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## EDUCATION

### Columbia University

New York, USA

**M.S. in Computer Science (Machine Learning concentration); GPA: 3.9/4.0**

Aug 2018 - Dec 2019

Relevant Coursework - Computer Vision, Applied Deep Learning, Machine Learning, Natural Language Processing

- Awarded CA Fellowship and tuition waiver (Graduate Course Assistant, Databases, [Prof. Donald Ferguson](#))

### National Institute of Technology Nagpur

Nagpur, India

**B.Tech. in Electronics and Communication Engineering; GPA: 8.21/10.0**

Jul 2012 - May 2016

## EXPERIENCE

### Living Things.AI

Mumbai, India

**Senior Data Scientist** | Python, scikit-learn, SQL, Quicksight, LLMs

Aug 2023 - Present

- Led client engagements, converting pilots into revenue contracts with low-churn probability (3x clients, 2x revenue)
- Optimized HVAC performance for commercial sites, cutting electricity costs by ~30% and reducing carbon footprint
- Engineered features for parameter recommendations and designed KPIs to drive operational efficiency improvements
- Developing regression models to predict baseline consumption and estimate savings under variable weather conditions

### Microsoft

Redmond, USA

**Applied AI & Data Scientist II**, Azure AI | Python, Pandas, Spark, scikit-learn, Power BI, Databricks

Mar 2020 - May 2023

- Built end-to-end anomaly detection pipelines to prevent high-severity outages and save over \$25 Million in operations
- Developed statistical/ML models for time series data using EDA, hypothesis testing, and feature engineering
- Increased average precision/recall to over 0.8 by incorporating domain knowledge, text parsing, and flexible alert criteria
- Designed a scalable framework to improve data quality and reduce time to onboard new scenarios by 50%

### PayPal

Scottsdale, USA

**Machine Learning Intern**, SRE-AI & Innovation Lab | Python, scikit-learn, Pandas, Gensim

Jun 2019 - Aug 2019

- Designed ~50 experiments with text features, PCA visualizations, and clustering algorithms to find optimal cluster count
- Clustered incident tickets to detect similarity, reduce mitigation time, and save an estimated \$1.2 Million in operations

### Citigroup

Pune, India

**Software Developer**, Citi Private Bank, and Compliance Technology | Java, Spring, SQL

Jul 2016 - Jun 2018

- Developed modules for access management of accounts and relationships requested by high-net-worth clients
- Designed the real-time data migration platform for CitiCode, an application to minimize regulatory risks

### Google Summer of Code

Open Source, Remote

**Student Developer**, Anita Borg Institute | Python, Django, Selenium

May 2016 - Aug 2016

- Increased test coverage from 0 to 83%. Boosted scalability by migrating the architecture to Page Object Model design
- Reduced testing time by 45% by minimizing object duplication. Trusted with repository merging rights upon completion

## SELECTED PROJECTS

**Detecting Cancer Metastasis in Lymph Nodes** | Python, TensorFlow

Mar 2019 - May 2019

- Trained a CNN to detect cancerous cells from Gigapixel slides. Modeled pathologist action via parallel inception towers
- Obtained an average recall score of 0.848 and F1 score of 0.81 for test slides in the CAMELYON16 Dataset

**Predicting transitions for Dependency Parsing** | Python, Keras, TensorFlow

Nov 2018 - Dec 2018

- Designed a feed-forward neural network to predict the transitions of an arc-standard dependency parser
- Implemented a Greedy Parsing algorithm to decode sentences based on predicted transitions. Obtained a LAS ~ 70

## SKILLS

**Languages:** Python, SQL

**Libraries:** LangChain, OpenAI, AWS, TensorFlow, scikit-learn, Pandas

## EXTRACURRICULAR / OPEN SOURCE

**Google Summer of Code 2017 Mentor, GNU Mailman** - Evaluated applications and mentored the student for 4 months

**Google Code-In 2016 Mentor** - Created ~30 open-source coding tasks and helped high school students contribute them